

## CAP WHITTLE, QC, Canada

WMO: 713740

Lat: 50.177N

Lon: 60.121W

Elev: 7

StdP: 101.24

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -21.7      | -19.4 | -27.7                         | 0.3 | -18.9 | -24.5 | 0.4 | -18.9 | 22.1                  | -5.5 | 20.4 | -4.8 | 7.9                      | 300  | 0.978 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.3                    | 16.8            | 14.7 | 16.1 | 14.4 | 15.4 | 14.0 | 15.5                | 16.0 | 14.9 | 15.4 | 14.1 | 14.7 | 6.4                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 15.3                            | 10.9 | 15.7 | 14.6 | 10.4 | 15.1 | 13.7 | 9.8 | 14.5 | 43.5          | 16.1 | 41.7 | 15.4 | 39.5 | 14.7 | 18.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| (4) 19.1          | 17.1 | 15.4 | -24.9                      | 19.5 | 2.3                | 1.4 | -26.6                                              | 20.5 | -27.9      | 21.4 | -29.2      | 22.2 | -30.9      | 23.2 |  |  |
| (5)               |      |      | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.2    | -9.2  | -9.1  | -5.3  | -0.7  | 3.7   | 8.1   | 11.8  | 13.5  | 10.5  | 6.1   | 1.0   | -5.1 |
|                                                                                  | DBStd       | 8.47   | 5.59  | 5.21  | 4.41  | 2.33  | 2.09  | 2.07  | 1.80  | 1.92  | 2.21  | 2.32  | 3.46  | 4.88 |
|                                                                                  | HDD10.0     | 3069   | 597   | 534   | 474   | 320   | 197   | 63    | 4     | 2     | 19    | 121   | 269   | 468  |
|                                                                                  | HDD18.3     | 5897   | 855   | 768   | 733   | 570   | 455   | 307   | 203   | 149   | 234   | 378   | 519   | 727  |
|                                                                                  | CDD10.0     | 213    | 0     | 0     | 0     | 0     | 0     | 6     | 60    | 111   | 36    | 1     | 0     | 0    |
|                                                                                  | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.6    | 8.8   | 8.5   | 8.4   | 7.4   | 6.6   | 6.2   | 5.9   | 6.0   | 7.3   | 7.8   | 8.8   | 9.2  |
| Precipitation                                                                    | PrecAvg     | 987    | 75    | 63    | 69    | 65    | 76    | 79    | 101   | 100   | 91    | 92    | 92    | 83   |
|                                                                                  | PrecMax     | 1229   | 124   | 111   | 133   | 95    | 160   | 139   | 178   | 171   | 149   | 155   | 145   | 141  |
|                                                                                  | PrecMin     | 798    | 30    | 28    | 34    | 37    | 29    | 34    | 61    | 36    | 42    | 42    | 42    | 45   |
|                                                                                  | PrecStd     | 90     | 22    | 21    | 22    | 17    | 30    | 23    | 27    | 32    | 26    | 25    | 26    | 24   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 3.2   | 2.5   | 3.3   | 4.8   | 10.8  | 14.8  | 17.6  | 18.4  | 16.3  | 12.1  | 8.7   | 5.0  |
|                                                                                  |             | MCWB   | 2.5   | 2.2   | 2.3   | 2.0   | 8.0   | 10.5  | 14.0  | 15.3  | 13.5  | 10.9  | 8.2   | 4.4  |
|                                                                                  | 2%          | DB     | 1.5   | 0.8   | 1.9   | 3.5   | 8.7   | 13.0  | 16.1  | 17.3  | 15.3  | 10.7  | 7.4   | 3.6  |
|                                                                                  |             | MCWB   | 1.0   | 0.3   | 0.7   | 1.7   | 6.6   | 10.0  | 13.4  | 15.2  | 13.0  | 9.5   | 6.7   | 3.0  |
|                                                                                  | 5%          | DB     | 0.1   | -0.5  | 0.7   | 2.7   | 7.5   | 11.8  | 15.2  | 16.6  | 14.5  | 10.0  | 6.4   | 2.6  |
|                                                                                  |             | MCWB   | -0.7  | -0.9  | -0.2  | 1.3   | 5.8   | 9.8   | 13.3  | 15.0  | 12.5  | 8.9   | 5.6   | 1.9  |
|                                                                                  | 10%         | DB     | -1.2  | -1.6  | -0.3  | 1.9   | 6.4   | 10.8  | 14.3  | 16.0  | 13.6  | 9.3   | 5.5   | 1.5  |
|                                                                                  |             | MCWB   | -1.9  | -2.2  | -1.0  | 0.8   | 5.1   | 9.5   | 12.9  | 14.7  | 12.0  | 8.3   | 4.6   | 0.8  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 2.9   | 2.5   | 3.1   | 3.2   | 9.1   | 11.7  | 15.6  | 16.6  | 14.1  | 11.3  | 8.2   | 4.5  |
|                                                                                  |             | MCDB   | 3.0   | 2.5   | 3.2   | 4.1   | 10.3  | 12.6  | 16.4  | 17.0  | 14.5  | 11.8  | 8.3   | 4.8  |
|                                                                                  | 2%          | WB     | 1.2   | 0.2   | 1.1   | 2.2   | 7.5   | 10.9  | 14.7  | 16.0  | 13.1  | 9.9   | 6.7   | 3.3  |
|                                                                                  |             | MCDB   | 1.6   | 0.3   | 1.7   | 2.8   | 8.6   | 11.6  | 15.3  | 16.4  | 13.8  | 10.4  | 7.0   | 3.7  |
|                                                                                  | 5%          | WB     | -0.7  | -1.2  | 0.1   | 1.6   | 6.3   | 10.4  | 13.9  | 15.4  | 12.4  | 9.2   | 5.7   | 2.3  |
|                                                                                  |             | MCDB   | -0.3  | -0.9  | 0.7   | 2.2   | 7.4   | 11.1  | 14.5  | 15.8  | 13.2  | 9.8   | 6.1   | 2.8  |
|                                                                                  | 10%         | WB     | -2.1  | -2.9  | -0.7  | 1.0   | 5.5   | 9.8   | 13.2  | 14.9  | 11.8  | 8.5   | 4.7   | 1.1  |
|                                                                                  |             | MCDB   | -1.7  | -2.4  | -0.2  | 1.6   | 6.4   | 10.6  | 13.9  | 15.3  | 12.7  | 9.1   | 5.3   | 1.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.8   | 6.4   | 5.7   | 3.4   | 3.3   | 3.4   | 3.3   | 3.3   | 3.3   | 3.1   | 3.9   | 5.5  |
|                                                                                  |             | MCDBR  | 7.0   | 5.9   | 5.5   | 3.8   | 5.0   | 4.9   | 4.6   | 3.9   | 3.8   | 3.4   | 4.1   | 5.4  |
|                                                                                  |             | MCWBR  | 7.1   | 6.9   | 5.9   | 3.1   | 3.3   | 3.3   | 3.0   | 3.2   | 3.6   | 3.3   | 4.5   | 5.4  |
|                                                                                  | 5% WB       | MCDBR  | 6.9   | 6.8   | 5.7   | 3.3   | 4.1   | 3.9   | 3.6   | 3.6   | 3.3   | 3.1   | 4.1   | 5.3  |
|                                                                                  |             | MCWBR  | 7.1   | 7.0   | 5.7   | 2.9   | 3.2   | 3.1   | 3.0   | 2.9   | 3.7   | 3.2   | 4.7   | 5.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.282  | 0.292 | 0.304 | 0.338 | 0.362 | 0.380 | 0.412 | 0.384 | 0.354 | 0.334 | 0.300 | 0.282 |      |
|                                                                                  | taud        | 2.388  | 2.342 | 2.367 | 2.396 | 2.426 | 2.394 | 2.357 | 2.435 | 2.492 | 2.501 | 2.513 | 2.415 |      |
|                                                                                  | Ebn at Noon | 766    | 851   | 906   | 907   | 895   | 878   | 842   | 850   | 843   | 796   | 748   | 712   |      |
|                                                                                  | Edh at Noon | 63     | 84    | 98    | 107   | 109   | 114   | 116   | 103   | 88    | 73    | 56    | 53    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.97   | 1.79  | 3.12  | 4.33  | 5.11  | 5.37  | 5.13  | 4.51  | 3.38  | 2.10  | 1.07  | 0.74  |      |
|                                                                                  | RadStd      | 0.10   | 0.18  | 0.32  | 0.47  | 0.37  | 0.53  | 0.47  | 0.25  | 0.23  | 0.15  | 0.10  | 0.08  |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (5 neighbors) |  | N/A   | N/A     | +1.53  | N/A     | +0.41 | +0.43 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page